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OLED DISPLAY PANEL AND DISPLAY DEVICE

Abstract

The present invention provides an organic light emitting diode (OLED) display panel. At least one light transmissive region used for placing photosensitive elements thereon is defined in a display region of the OLED display panel. Special-shaped sub-pixels are arranged at a periphery of each of the at least one light transmissive region. Each special-shaped sub-pixel includes at least one recess, and the at least one recess is at one side of the special-shaped sub-pixel facing a corresponding one of the at least one light transmissive region. An edge of each of the at least one recess conforms in shape to an edge of a corresponding one of the at least one light transmissive region.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of U.S. patent application Ser. No. 17/933,481, filed on Sep. 20, 2022, which is a continuation of U.S. patent application Ser. No. 16/496,422, filed on Sep. 21, 2019, which is a National Phase entry of PCT Patent Application No. PCT/CN2019/077247, filed on Mar. 7, 2019, which claims priority to Chinese Patent Application No. 201811640213.5, filed on Dec. 29, 2018, with the National Intellectual Property Administration of China. This application further claims priority to Chinese Patent Application No. 202210766034.6, filed on Dec. 29, 2018, with the National Intellectual Property Administration of China. The disclosures of the aforementioned applications are hereby incorporated by reference in their entireties.

1. FIELD OF DISCLOSURE

[0002] The present invention relates to a field of display devices and in particular, to an organic light emitting diode (OLED) display panel and a display device.

2. DESCRIPTION OF RELATED ART

[0003] Organic light emitting diode (OLED) displays are becoming mainstream in the display industry due to their wide viewing angles, high color gamut, and low power consumption. Industries aim to develop flexible and foldable OLED displays.

[0004] Nowadays, full-screen mobile phones have become mainstream on the market. While increasing a screen ratio, a space of a sensing region is also decreased. In order to obtain an optimal configuration of the sensing region, sensing elements of the sensing region are usually placed under a display panel. The display panel is provided with an opening which matches the sensing region, and the sensing elements are disposed in the opening. The sensing region is provided with a plurality of light transmissive holes for exposing photosensitive lenses, but this way, the opening at an edge of the display panel cannot display images, and the opening causes a decrease in a quantity of pixels in the display panel and lowers image quality of the display panel.

[0005] In summary, in conventional OLED display panels, setting a sensing region needs to sacrifice a large number of pixels, thereby reducing a display area of the display panel and reducing image quality of the display panel. Therefore, it is necessary to solve the existing problems in conventional techniques.

SUMMARY

[0006] The present invention provides an OLED display panel, in which pixels are disposed in a sensing region and outside light transmissive holes of the display panel, thereby increasing a display area of the display panel. The present invention solves existing problems in conventional display panels. The problems in the conventional display panels are that having a sensing region needs to sacrifice a large number of pixels, thereby reducing the display area of the display panel, and reducing image quality of the display panel.

[0007] Accordingly, the present invention provides solutions as follows.

[0008] The present invention provides an organic light emitting diode (OLED) display panel, including: [0009] a display region; [0010] at least one light transmissive region defined in the display region; [0011] a plurality of special-shaped sub-pixels disposed at a periphery of each of the at least one light transmissive region; [0012] wherein each of the special-shaped sub-pixels includes, at one side facing a corresponding one of the at least one light transmissive region, at least one recess, and an edge of each of the at least one recess conforms in shape to an edge of a corresponding one of the at least one light transmissive region.

[0013] According to one embodiment of the present invention, each of the at least one light transmissive region includes at least one curved edge protruding outwardly, and each of the special-shaped sub-pixels includes, at one side facing a corresponding one of the at least one curved edge, one curved recess.

[0014] According to one embodiment of the present invention, each of the at least one light transmissive region has a symmetrical shape about its center.

[0015] According to one embodiment of the present invention, a shape of each of the light transmissive regions is a circular shape or a parallelogram with rounded corners.

[0016] According to one embodiment of the present invention, an area of each of the special-shaped sub-pixels is equal to an area of each of a plurality of regular-shaped sub-pixels of the display region.

[0017] According to one embodiment of the present invention, the special-shaped sub-pixels surround the at least one light transmissive region, and the special-shaped sub-pixels are arranged along virtual horizontally extending lines and virtual perpendicularly extending lines connecting adjacent ones of the regular-shaped sub-pixels.

[0018] According to one embodiment of the present invention, each of the at least one light transmissive region has a circular shape, and the special-shaped sub-pixels are arranged at respective four ends of two orthogonal diameters of each of the at least one light transmissive region.

[0019] According to one embodiment of the present invention, at each of the four ends of the two orthogonal diameters is disposed one of the special-shaped sub-pixels, an arc length of one of the at least one recess of each special-shaped sub-pixel is substantially equal to one-fourth of a circumference of a corresponding one of the at least one light transmissive region.

[0020] According to one embodiment of the present invention, each of the at least one light transmissive region is a parallelogram with four rounded corners, and the special-shaped sub-pixels are disposed at respective ends of two diagonals of each of the at least one light transmissive region, or each of the special-shaped sub-pixels is disposed between each two adjacent ones of four ends of two diagonals of each of the at least one light transmissive region.

[0021] According to one embodiment of the present invention, at four diagonal ends or each two adjacent ones of the four diagonal ends of each of the at least one light transmissive region is disposed one of the special-shaped sub-pixels, and an arc length of one of the at least one recess of each special-shaped sub-pixel is substantially equal to one-fourth of a circumference of a corresponding one of the at least one light transmissive region.

[0022] According to one embodiment of the present invention, the display region includes two light transmissive regions disposed adjacent to each other, two special-shaped sub-pixels and one complementary-color sub-pixel between the two special-shaped sub-pixels are disposed on the same virtual horizontally extending line between the two light transmissive regions, and the complementary-color sub-pixel and the special-shaped sub-pixels at two sides thereof are of different colors.

[0023] According to one embodiment of the present invention, two of the special-shaped sub-pixels are disposed at each of four ends of the two orthogonal diameters, and an arc length of one of the at least one recess of each special-shaped sub-pixel is substantially one-twelfth of a circumference of

a corresponding one of the at least one light transmissive region.

[0024] According to one embodiment of the present invention, a complementary-color sub-pixel is disposed between the two special-shaped sub-pixels at one end of the same diameter, and the complementary-color sub-pixel and the two special-shaped sub-pixels at two sides thereof are of different colors.

[0025] According to one embodiment of the present invention, the present invention provides a display device, including the OLED display panel of claim 1, the OLED display panel including: [0026] the display region; [0027] a sensing region positioned in the display region; [0028] the at least one transmissive region positioned in the sensing region; and [0029] at least one photosensitive element disposed on a back side of the sensing region, the at least one photosensitive element being disposed corresponding to the at least one light transmissive region.

[0030] Advantages: The invention provides an OLED display panel, which expands an area of the pixels into the sensing region, and allows the pixels at the peripheries of the light transmissive regions to have ideal shapes, thereby increasing a display area of the display panel, improving the image quality of the display panel, and realizing a full screen. The present invention solves the existing problems in the conventional OLED display panel. Those problems are that a sensing region needs to sacrifice a large number of pixels, a display area of the conventional display panel is thus reduced, and image quality of the display panel is compromised.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0031] In order to more clearly illustrate the embodiments of the present disclosure or related art, figures which will be described in the embodiments are briefly introduced hereinafter. It is obvious that the drawings are merely for the purposes of illustrating some embodiments of the present disclosure, and a person having ordinary skill in this field can obtain other figures according to these figures without an inventive work or paying the premise.

[0032] FIG. 1 is a schematic structural view illustrating an organic light emitting diode (OLED) display panel according to the present invention;

[0033] FIGS. 2A to 2E are schematic views illustrating a pixel structure according to a first embodiment of the present invention; and

[0034] FIG. 3 is a schematic view illustrating the pixel structure according to a second embodiment of the present invention.

DETAILED DESCRIPTION OF EMBODIMENTS

[0035] Embodiments of the present disclosure are described in detail with reference to the accompanying drawings as follows. Directional terms such as up/down, front/rear, right/left, inside/outside, and the like may be used for the purpose of enhancing a reader's understanding about the accompanying drawings, but are not intended to be limiting. Specifically, the terminologies in the embodiments of the present disclosure are merely for the purpose of describing certain embodiments, but not intended to limit the scope of the invention. The same reference numbers are used throughout the drawings to refer to the same or similar parts.

[0036] The present invention is directed to solving existing problems in conventional organic light emitting diode (OLED) display panels. The problems are that a sensing region needs to sacrifice a large number of pixels, thereby reducing a display area of the display panel and lowering image quality of display panel. The present invention can solve the problems.

[0037] Referring to FIG. 1, the present invention provides an organic light emitting diode (OLED) display panel **101**. The OLED display panel **101** includes a display region **102**. A sensing region **103** positioned at an edge of the display region **102**. At least one light transmissive region **105** is positioned in the sensing region **103**. A sensor member is disposed on a back side of the sensing

region **103**. A photosensitive unit of the sensor member is disposed corresponding to the at least one light transmissive region **105**. The at least one light transmissive region **105** is concentrated within multiple circular regions **104** of the sensing region **103**. Sub-pixels are arrayed in the display region **102**, the sensing region **103**, and the circular regions **104**, so that the display region **102**, the sensing region **103**, and the circular region **104** of the OLED display panel **101** can display images. [0038] The sub-pixels positioned in the circular regions **104** are special-shaped sub-pixels **106** and are disposed at a periphery of each of the at least one light transmissive region **105**. An edge **1061** of each of the special-shaped sub-pixels **106** conforms in shape to an edge **1051** of a corresponding one of the at least one light transmissive region **105**. In other words, the edge **1061** of each of the special-shaped sub-pixels **106** has the same bending direction as the edge **1051** of a corresponding one of the at least one light transmissive region **105**, and the bending curvature/bending angle is the same or similar. The edge **1061** of each special-shaped sub-pixel **106** is spaced apart from the edge **1051** of the corresponding light transmissive region **105** by a predetermined distance. The special-shaped sub-pixels **106** match the light transmissive regions **105**, so that the sub-pixels have high conformity to the light transmissive regions **105**, and a blank area at the edge **1051** of each light transmissive region **105** is filled to form a pixel, thereby optimally eliminating an image sawtooth at the edge **1051** of each light transmissive region **105**, so that a displayed image of each circular region and a display image of other region are unified.

[0039] Each of the at least one light transmissive region **105** includes at least one curved edge protruding outwardly, and each of the special-shaped sub-pixels **106** includes one curved recess conforms in shape to the curved edge. An arc curvature of the recess of each special-shaped sub-pixel **106** is the same as or similar to an arc curvature of the curved edge of a corresponding one of the at least one light transmissive region **105**, and the recess is disposed at one side of each special-shaped sub-pixel **106** facing the light transmissive region **105**.

[0040] Each special-shaped sub-pixel **106** is different in shape from other sub-pixels in the sensing region **103** and the display region **102**, but has the same area, so that a displayed image at a periphery of each of the at least one light transmissive region **105** and displayed images of other regions can be unified.

First Embodiment

[0041] Referring to FIG. 2a, at least one light transmissive region **201** is disposed in the sensing region of the OLED display panel. A plurality of special-shaped sub-pixels **202** are disposed at the periphery of each of the at least one light transmissive region **201**. Each of the special-shaped sub-pixels **202** includes, at one side facing a corresponding one of the at least one light transmissive region **201**, at least one recess **203**.

[0042] A contour of each of the at least one light transmissive region **201** is a circle formed by a closed arc. Around each of the at least one light transmissive region **201** is symmetrically distributed the special-shaped sub-pixels **202**, and a shape of each of the at least one special-shaped sub-pixel **202** is approximately a right triangle. An oblique side of the right triangle is a circular arc to form the recess **203**, and the oblique side is arranged facing the light transmissive region **201**.

[0043] Two orthogonal (mutually perpendicular) diameters of each of the at least one light transmissive region **201** have four ends. Each of the special-shaped sub-pixels **202** is arranged corresponding to each of the four ends of the two orthogonal diameters of each of the at least one light transmissive region **201**. Meanwhile, the special-shaped sub-pixels **202** are also required to satisfy another positional condition as follows. Please also refer to FIG. 3, regular-shaped sub-pixels **303** are disposed in arrays in the display region and the sensing region, and a shape of each of the regular-shaped sub-pixels **303** is a standard planar geometric shape. Virtual horizontally extending lines **3031** and virtual perpendicularly extending lines **3032** connecting the regular-shaped sub-pixels **303** are extended to the periphery of each of the at least one light transmissive region **201**, the special-shaped sub-pixels **202** are located at intersections of the virtual horizontally extending lines **3031** and the virtual perpendicularly extending lines **3032**.

[0044] In a case where an area of each of the special-shaped sub-pixels **202** is the same as an area of each of the regular-shaped sub-pixels in the display region and the sensing region, an arc length of an inner arc of each of the special-shaped sub-pixels **202** is approximately equal to one-fourth of a circumference of a corresponding one of the at least one light transmissive region **201**, so that the special-shaped sub-pixels **202** surrounds the at least one light-transmissive region **201** to a greatest extent.

[0045] The sensing region is provided with the at least one light transmissive region **201**.

[0046] When the sensing region is provided with one of the at least one light transmissive region **201**, two rows of the special-shaped sub-pixels **202** are disposed around this light transmissive region **201**. Each row of the special-shaped sub-pixels **202** is matched up with a complementary-color sub-pixel **204**. The complementary-color sub-pixel **204** and the special-shaped sub-pixels **202** at the same row are different in color, so that the sub-pixels at the same row constitute one pixel unit.

[0047] For example, in the same row of sub-pixels, the complementary-color sub-pixel **204** is of a blue (B) color, the special-shaped sub-pixel **202** near the complementary-color sub-pixel **204** is of a red (R) color, and the special-shaped sub-pixel **202** adjacent to the red (R) special-shaped sub-pixel **202** is of a green (G) color.

[0048] As shown in FIG. 2B, when the sensing region is provided with two light transmissive regions, for example, the sensing region is provided with a first light transmissive region **2011** and a second light transmissive region **2012**, and the first light transmissive region **2011** and the first light transmissive region **2012** are positioned on the same horizontal line, have a same aperture diameter, and are disposed adjacent to each other. Two rows of sub-pixels are distributed around the first light transmissive region **2011** and the second light transmissive region **2012**. Each row of sub-pixels includes two of the special-shaped sub-pixels **202** located between the first light transmissive region **2011** and the second light transmissive region **2012**, a complementary-color sub-pixel **204** between the two special-shaped sub-pixels **202**, the special-shaped sub-pixel **202** at the other side of the first light transmissive region **2011**, and the special-shaped sub-pixel **202** at the other side of the second light transmissive region **2012**. The complementary sub-pixel **204** and the special-shaped sub-pixels **202** at two sides thereof are different in color, and the special-shaped sub-pixels **202** adjacent to each other in the same row are also different in color. Accordingly, in the same row of the sub-pixels, the complementary sub-pixel **204** and the special-shaped sub-pixels **202** at two sides thereof constitute one pixel unit, or the complementary-color sub-pixel **204** and the special-shaped sub-pixels **202** on either side thereof constitute a pixel unit.

[0049] For example, in the same row of the sub-pixels, the complementary-color sub-pixel **204** is green (G), the special-shaped sub-pixel **202** at one side of the complementary-color sub-pixel **204** is red (R), and the special-shaped sub-pixel **202** at the other side of the complementary-color sub-pixel **204** is blue (B), the special-shaped sub-pixel **202** adjacent to the red special-shaped sub-pixel **202** is blue (B), and the special-shaped sub-pixel **202** adjacent to the blue special-shaped sub-pixel **202** is red (R).

[0050] For example, the complementary-color sub-pixel **204** is of a rectangular shape.

[0051] As shown in FIG. 2C, the sensing region of the OLED display panel is provided with a first light transmissive region **2011** and a second light transmissive region **2012**, and the first light transmissive region **2011** and the second light transmissive region **2012** are positioned on the same horizontal line, have a same aperture diameter, and are disposed adjacent to each other. Two rows of the special-shaped sub-pixels **202** are distributed around the light transmissive region **2011** and the second light transmissive region **2012**, and the complementary-color sub-pixel **204** is disposed in each row of the special-shaped sub-pixels **202**.

[0052] FIG. 2C differs from FIGS. 2A and 2B in that a shape of the light transmissive region **201** is a parallelogram with rounded corners, and in detail, the shape of the light transmissive region **201** is a diamond with rounded corners.

[0053] The two diagonals of each light transmissive region **201** include four ends, and each of the special-shaped sub-pixels **202** is distributed between each adjacent two ends of the diagonals. An arc length of the recess **203** of each special-shaped sub-pixel **202** is approximately equal to one-fourth of a circumference of the corresponding light transmissive region **201**.

[0054] The shape of each special-shaped sub-pixel **202**, the position of each complementary-color sub-pixel **204**, and a color distribution thereof are the same as those of FIG. 2A, so a detailed description is not repeated herein for brevity.

[0055] As shown in FIG. 2D, the sensing region of the OLED display panel is provided with a first light transmissive region **2011** and a second light transmissive region **2012**, and the first light transmissive region **2011** and the second light transmissive region **2012** are positioned on the same horizontal line. The first light transmissive region **2011** and the second light transmissive region **2012** have a same aperture diameter and are adjacent to each other. Two rows of the special-shaped sub-pixels **202** are distributed around the first light transmissive region **2011** and the second light transmissive region **2012**. A complementary-color sub-pixel **204** is disposed in each row of the special-shaped sub-pixels **202**.

[0056] FIG. 2D is different from FIGS. 2A and 2B in that the shape of the light transmissive region **201** is a parallelogram with rounded corners, and in detail, the shape of the light transmissive region **201** is a square with rounded corners.

[0057] The two diagonals of each light transmissive region **201** have four ends, and each of the special-shaped sub-pixels **202** is distributed at each of the four ends of the two diagonals. An arc length of the recess **203** of each special-shaped sub-pixel **202** is approximately equal to one-fourth of a circumference of the corresponding light transmissive region **201**.

[0058] The shape of each special-shaped sub-pixel **202**, the position of each complementary-color sub-pixel **204**, and a color distribution thereof are the same as those of FIG. 2A, so a detailed description is not repeated herein for brevity.

[0059] As shown in FIG. 2E, the sensing region of the OLED display panel is provided with the at least one light transmissive region **201**, and the special-shaped sub-pixels **202** are distributed around the at least one light transmissive region **201**.

[0060] Each of the at least one light transmissive region **201** has a circular shape. Two orthogonal diameters of each of the at least one light transmissive region **201** have four ends. One of the special-shaped sub-pixels **202** is arranged at each of the four ends of the two orthogonal diameters.

[0061] The shape of each special-shaped sub-pixel **202** is similar to an arch shape, and the special-shaped sub-pixel **202** includes an inner arc and an outer arc. The inner arc is located on one side of each special-shaped sub-pixel **202** facing the light transmissive region **201**. The outer arc is located on one side of each special-shaped sub-pixel **202** away from the light transmissive region **201**. An arc length of the inner arc of each special-shaped sub-pixel **202** is approximately equal to one-fourth of a circumference of the light transmissive region **201**.

[0062] In a case where a ratio of the inner arc of each special-shaped sub-pixel **202** to the circumference of each of the light transmissive regions **201** of different aperture diameters is kept the same, the shape of each special-shaped sub-pixel **202** is changed by changing a curvature of the outer arc. This way, an area of each special-shaped sub-pixel **202** can be kept the same as an area of each regular-shaped sub-pixel. In the drawing, broken lines show the downsized light transmissive region and the deformed special-shaped sub-pixels.

Second Embodiment

[0063] Referring to FIG. 3, the sensing region of the OLED display panel is provided with regular-shaped sub-pixels **303**, light transmissive regions **301**, and special-shaped sub-pixels **306** located at peripheries of the light transmissive regions **301**. Each special-shaped sub-pixel **302** is provided with a recess on at least one side facing a corresponding one of the light transmissive regions **301**.

[0064] Each light transmissive region **301** has a circular shape formed by a closed arc, and the special-shaped sub-pixels **302** is symmetrically distributed around the light transmissive regions

301. Two orthogonal (mutually perpendicular) diameters of the light transmissive region **301** have four ends, and two of the special-shaped sub-pixels **302** are disposed at positions corresponding to each of the four ends. An arc length of the recess of each special-shaped sub-pixel **302** is approximately equal to 1/12 of a circumference of a corresponding one of the light transmissive regions **301**.

[0065] A complementary-color sub-pixel **304** is disposed between the two special-shaped sub-pixels **302** at one end of the same diameter, and the complementary-color sub-pixel **304** and the two special-shaped sub-pixels **302** at two sides thereof are of different colors to thereby constitute a pixel unit **305**.

[0066] A shape of each special-shaped sub-pixel **302** is similar to an equilateral triangle, and the three sides of the special-shaped sub-pixel **302** are disposed with the three recesses respectively for matching the light transmissive regions **301** which the recesses face respectively. At least one side of the special-shaped sub-pixel **302** matches the complementary-color sub-pixel **304**, and the complementary-color sub-pixel **304** has a circular shape.

[0067] For example, the sensing region is provided with four light transmissive regions **301** of a same aperture diameter, which includes a central light transmissive region **301**, and three light transmissive regions at three ends of diameters of the central light transmissive region **301**, a pixel unit **305** is arranged between adjacent two of the light transmissive regions **301**. Each of the pixel units **305** includes four of the special-shaped sub-pixels **302**, arranged in an array, and each complementary-color sub-pixel **304** is located at a center of each four of the special-shaped sub-pixels **302**. In the same row and in the same column of the sub-pixels, the adjacent special-shaped sub-pixels **302** are of different colors, and the special-shaped sub-pixels **302** and the complementary-color sub-pixel **304** are of different colors.

[0068] For example, the complementary-color sub-pixel **304** is blue (B), and the special-shaped sub-pixels **302** around each complementary-color sub-pixel **304** is red (R) and green (G) arranged alternately.

[0069] According to the above objectives of the present invention, the display device is provided, including the OLED display panel of the first embodiment and the second embodiment. The OLED display panel includes a display region, a sensing region located in the display region, and a light transmissive region. At least one photosensitive element is disposed on a back side of the sensing region, and a photosensitive unit of the photosensitive element is disposed corresponding to the light transmissive region.

[0070] Specific principles of the display device of the present invention are the same as or similar to the above descriptions in the preferable embodiments of the OLED display panel. For details, refer to the related descriptions in the preferable embodiments of the liquid crystal display panel, and a detailed description is not repeated herein for brevity.

[0071] The OLED display panel of the present invention expands an area of the pixels into the sensing region, allows the pixel units at the peripheries of the light transmissive regions to have an ideal shape, thereby increasing a display area of the display panel and improving image quality of the display panel. In conventional OLED display panels, setting a sensing region requires sacrificing a large number of pixels, which causes reduction in a display area of the conventional display panel and lowers image quality of the display panel.

[0072] It is to be understood that the above descriptions are merely the preferable embodiments of the present invention and are not intended to limit the scope of the present invention. Equivalent changes and modifications made in the spirit of the present invention are regarded as falling within the scope of the present invention.

Claims

- 1.** An organic light emitting diode (OLED) display panel, comprising: a display region, wherein a plurality of sub-pixels are arranged in an array in the display region; and a sensing region, wherein a plurality of pixel units and a plurality of light transmissive regions are arranged in an array in the sensing region, the light transmissive regions disposed among different pixel units; wherein each of the pixel units comprises special-shaped sub-pixels having a shape different from the sub-pixels in the display region, each of the special-shaped sub-pixels having a recessed side, the recessed side being disposed at one side of the special-shaped sub-pixel facing one of the light transmissive regions; and wherein the pixel units are arranged in an island chain distribution in the sensing region.
- 2.** The OLED display panel according to claim 1, wherein the pixel units further comprise a plurality of regular-shaped sub-pixels, and the special-shaped sub-pixels are disposed at two sides of each of the regular-shaped sub-pixels.
- 3.** The OLED display panel according to claim 2, wherein the regular-shaped sub-pixels are green sub-pixels, the special-shaped sub-pixels are red sub-pixels and blue sub-pixels, an area of each of the special-shaped sub-pixels in each of the pixel units is larger than an area of each of the regular-shaped sub-pixels, and each of the regular-shaped sub-pixels is rectangular or circular shaped.
- 4.** The OLED display panel according to claim 2, wherein a contour of each of the special-shaped sub-pixels comprises two mutually perpendicular sides and the recessed side, one of the two mutually perpendicular sides facing the regular-shaped sub-pixel.
- 5.** The OLED display panel according to claim 1, wherein each of the recessed sides is spaced from an edge of a corresponding one of the light transmissive regions.
- 6.** The OLED display panel according to claim 1, wherein each of the recessed sides conforms in shape to an edge of a corresponding one of the light transmissive regions, and each of the light transmissive regions has a centrally symmetrical shape.
- 7.** The OLED display panel according to claim 1, wherein a spacing between any two adjacent ones of the pixel units in the sensing region is equal, the pixel units are grouped in a square or diamond arrangement in the sensing region, and each of the light transmissive regions is located within a pixel unit group having the square or diamond arrangement.
- 8.** The OLED display panel according to claim 1, wherein a minimum pixel unit group comprises a combination of four of the pixel units, and the four pixel units form an enclosing pattern surrounding one of the light transmissive regions.
- 9.** The OLED display panel according to claim 2, wherein: an edge of each of the regular-shaped sub-pixels conforms in shape to an edge of an adjacent special-shaped sub-pixel; an edge of each of the special-shaped sub-pixels conforms in shape to an edge of an adjacent light transmissive region; and each of the pixel units in the sensing region comprises the regular-shaped sub-pixels and the special-shaped sub-pixels, wherein the regular-shaped sub-pixels and the special-shaped sub-pixels are arranged in rows or columns, and the adjacent pixel units share the special-shaped sub-pixels.
- 10.** The OLED display panel according to claim 2, wherein each of the pixel units in the sensing region comprises one regular-shaped sub-pixels and the special-shaped sub-pixels, wherein the regular-shaped sub-pixel is located at an inner side of the pixel unit, and the special-shaped sub-pixels are located at an outer side of the regular-shaped sub-pixels and are disposed close to one of the light transmissive regions.
- 11.** The OLED display panel according to claim 10, wherein each of the pixel units comprises sub-pixels of at least three colors, wherein one or more sub-pixels emit each of the at least three colors, and the sub-pixel located at the inner side of the pixel unit is different in color from the sub-pixels located on two sides of the pixel unit.
- 12.** The OLED display panel according to claim 11, wherein one or more sub-pixels of the same color are arranged diagonally with respect to the sub-pixel located at the inner side of the pixel unit.
- 13.** The OLED display panel according to claim 1, wherein a boundary pixel region is defined

between the display region and the sensing region, and the pixel units in the boundary pixel region are arranged in an oblique direction.

14. The OLED display panel according to claim 1, wherein: the pixel units are arranged in an array distribution in the sensing region, and in at least one of a row or column arrangement direction, the pixel units and the light transmissive regions are alternately disposed; and the pixel units in adjacent rows or columns are offset from each other.

15. The OLED display panel according to claim 14, wherein each of the pixel units comprises at least a red pixel, a green pixel, and a blue pixel, wherein the red pixel and the green pixel are located in the same row or column, and the blue pixel is located in a different row or column from the red pixel and the green pixel.

16. The OLED display panel according to claim 14, wherein the light transmissive regions have a horizontal arrangement direction, a vertical arrangement direction, and an inclined arrangement direction between the horizontal and vertical arrangement directions, the light transmissive regions in the horizontal or vertical arrangement direction are offset from each other, and the light transmissive regions in the inclined arrangement direction are continuously disposed.

17. The OLED display panel according to claim 16, wherein at least two of the pixel units are disposed around a periphery of each of the light transmissive regions, and the two pixel units are rotationally symmetrical with respect to a center of the light transmissive region.

18. The OLED display panel according to claim 16, wherein an opening shape of each of the light transmissive regions is a geometric shape, and the opening of at least some of the light transmissive regions is circular or a square with rounded corners.

19. The OLED display panel according to claim 18, wherein: within a same unit area, a number of the pixel units in the sensing region is less than a number of the pixel units in the display region; a spacing between adjacent pixel units in the sensing region is greater than a spacing between adjacent pixel units in the display region; an edge of the sensing region and an edge of the display region form a smooth transition; and the pixel units at the edge of the sensing region and the sub-pixels at the edge of the display region are arranged in an arc-shaped pattern.

20. A display device, comprising an OLED display panel, the OLED display panel comprising: a display region, wherein a plurality of sub-pixels are arranged in an array in the display region; a sensing region, wherein a plurality of pixel units and a plurality of light transmissive regions are arranged in an array in the sensing region, and the light transmissive regions are disposed among different pixel units; wherein each of the pixel units comprises special-shaped sub-pixels having a shape different from the sub-pixels in the display region, each of the special-shaped sub-pixels having a recessed side, the recessed side being disposed at one side of the special-shaped sub-pixel facing one of the light transmissive regions; and wherein the pixel units are arranged in an island chain distribution in the sensing region.
